AI Application Testing

Foreword

- This course describes the main process and methods of AI application testing.

Objectives

- Upon completion of this course, you will be able to:
  - Understand the theories and methods of AI application testing.
  - Understand the role of AI application testing in model iteration and optimization.

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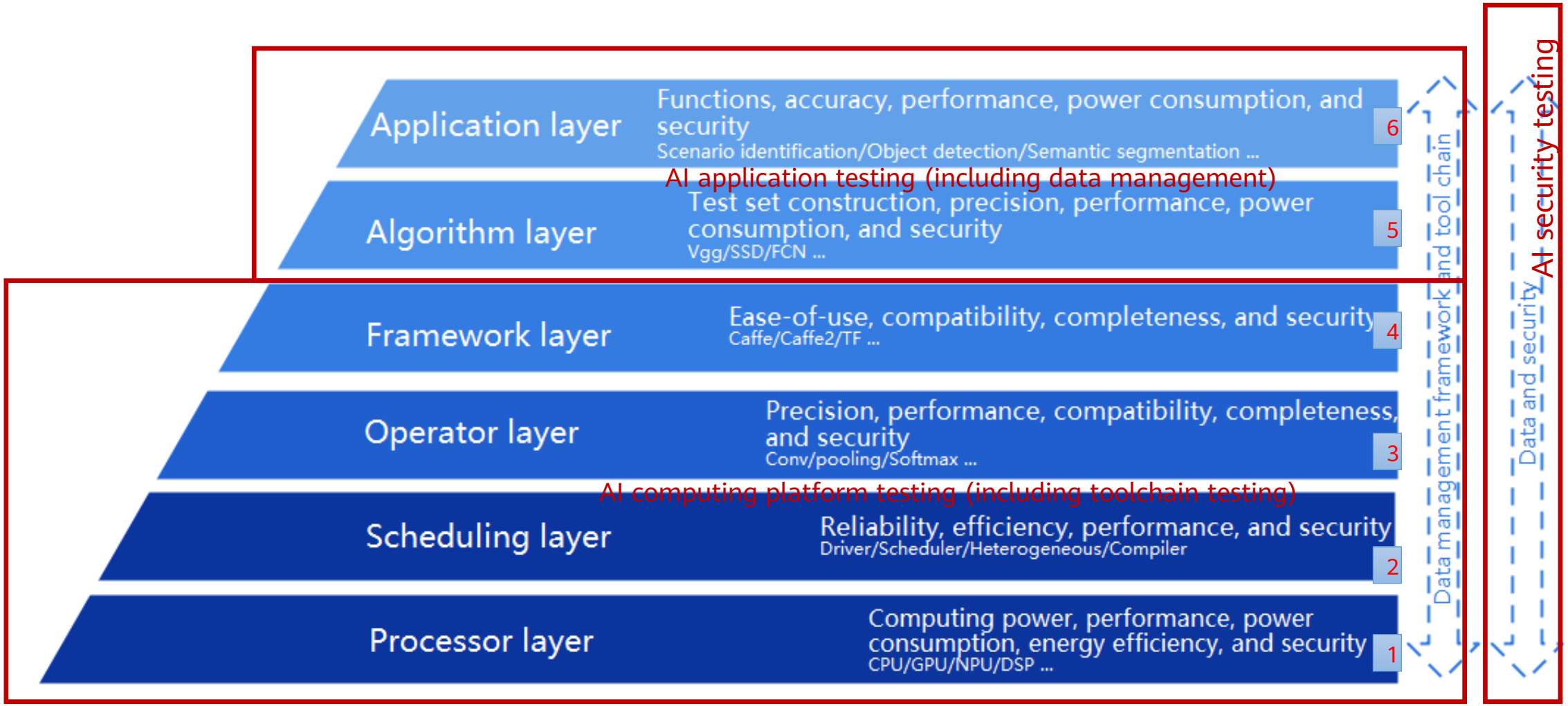
AI Test Task Layers



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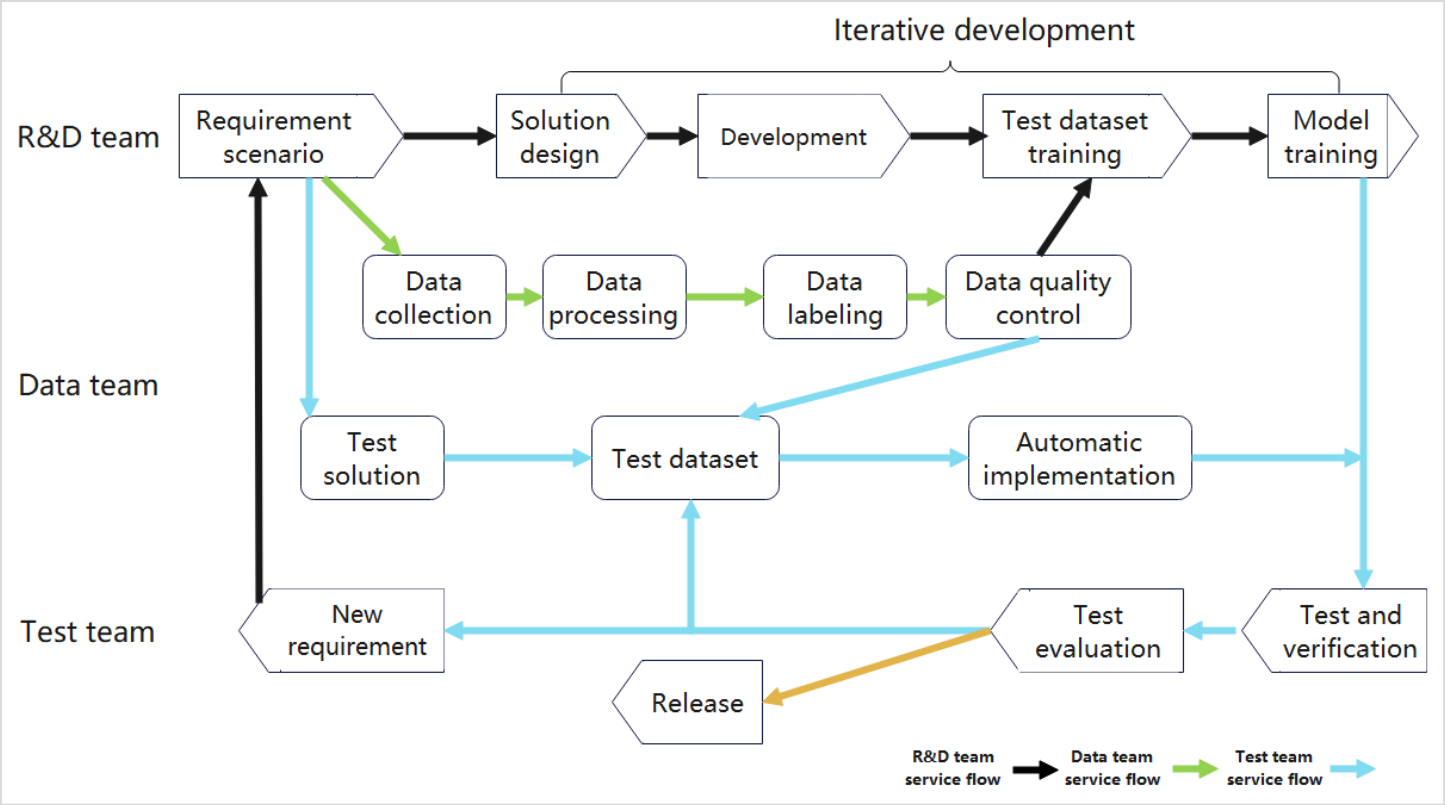
AI Application Test Framework



Key roles:
AI development engineers are responsible for design, development implementation, data acceptance, self-test, and optimization.
Data engineers are responsible for data collection, data cleaning, data integration, data labeling, and data quality check.
Test engineers are responsible for test data collection, test scenario analysis, test solutions, test cases, test strategies, test automation, test acceptance, and evaluation.

Test Process and Activities

Scenario analysis	Sort out typical application scenarios, subjects, and features based on the requirement description, and refine the scenarios.
Feature factor analysis	Based on scenario analysis, analyze feature factors that affect the identification rate of AI models and establish a feature factor library. Different AI scenarios have different requirements and complexity of feature factors. For example, feature factors in CV application scenarios include shooting factors (exposure and angle), environment factors (weather and background), and attribute factors (age and skin tone of a person).
Data construction	Build datasets based on application scenarios and feature factors, including but not limited to data types, factor proportion, environment factors, and specific requirements.
Evaluation metrics	Different service scenarios focus on different metrics, and the evaluation results are different. So the metrics and related definitions need to be determined based on the scenarios and focuses. For example, some services focus on recall for comprehensive results. Some services focus on accuracy. Some services feature imbalanced classes, so they focus on accuracy of each class. Some services are sensitive to memory, so they focus on model size.
Automation solution	Automated batch testing and evaluation metric collection

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Feature Factor Analysis

- Feature factor analysis is to analyze the factors that affect the identification performance of AI models based on the input data. During AI development, data collection and processing are important prerequisites. **A model is only as good as the data it is fed.**
- During AI inference, good input data is also important. **Good outputs can be obtained only when good data is available.**

Example: Factor Library of Facial Features (1)

Factor Category	Factor Subcategory	Typical Factor	Typical Value
Shooting	Shooting	Pitch angle	Within 30 degrees (up and down), above 30 degrees (up and down)
		Rotation angle	Within 15, 30, 45, 60 degrees (left and right), above 60 degrees (left and right)
		Not focused	Slightly out of focus (perceivable), severely out of focus
		Exposure	Overexposure, underexposure
		Special effect	Nostalgic, black and white, whitening
		Color saturation	Oversaturated, undersaturated
		Motion blur	Smear, light painting
		Main body location	9 box grid
		Entity integrity	< 30%, 50%, 80%
		Jitter	Slight jitter, severe jitter, horizontal jitter, vertical jitter, tilt jitter
		Screen-to-body ratio	20%, 30%, 60%
		Focus	Macro and telephoto
		Noise	Add noise to the image.

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Example: Factor Library of Facial Features (2)

Factor Category	Factor Subcategory	Typical Factor	Typical Value
Environment	Environment	Time	Morning, afternoon, night
		Weather	Cloudy, sunny, rainy, snowy
		Background	Simple background, complex background
		Light	Front light, backlight
		Light source	Street lamp, point light, surface light
		Contrast	Obvious, not obvious
		Location	Indoor, outdoor

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Example: Factor Library of Facial Features (3)

Factor Category	Factor Subcategory	Typical Factor	Typical Value
Object	People	Gender	Male, female
		Age	Baby (0-1 year old), toddler (1-4 years old), child (5-11 years old), teen (12-18 years old), young (19-35 years old), adult (36-59 years old), aged (over 60 years old)
		Skin tone	Yellow, white, black
		Facial expression	Normal, happy, sad, surprised, angry, pouted lips, tricky face
		Wearing	Headwear, hat, sunglasses, glasses, mask, scarf, earwear
		Appearance	Hairstyle, beard, makeup
		Number of targets	1, 2, 3, multiple
		Angle	Front face, side face, upward, downward

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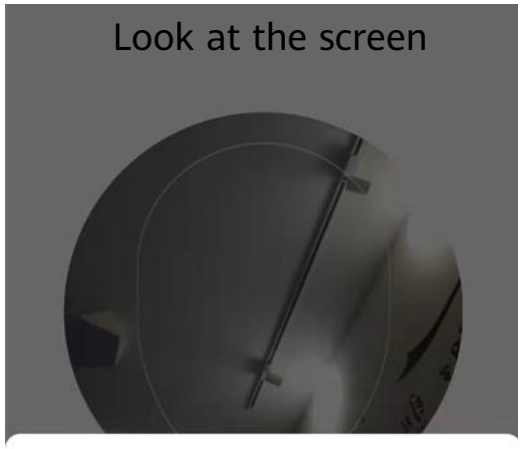
Constraints Based on Feature Factors

- In facial recognition scenarios, set constraints to guide users to take high-quality face images to improve the facial recognition rate.



When taking a photo, please note the following:

No glasses No masks Look straight ahead

Do not wear a mask

Keep your face in the specified area

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Feature Factor Examples

Overexposure ✗

Underexposure ✗

Blurry image ✗

Background interference ✗

Tilted image ✗

Ideal image ✓

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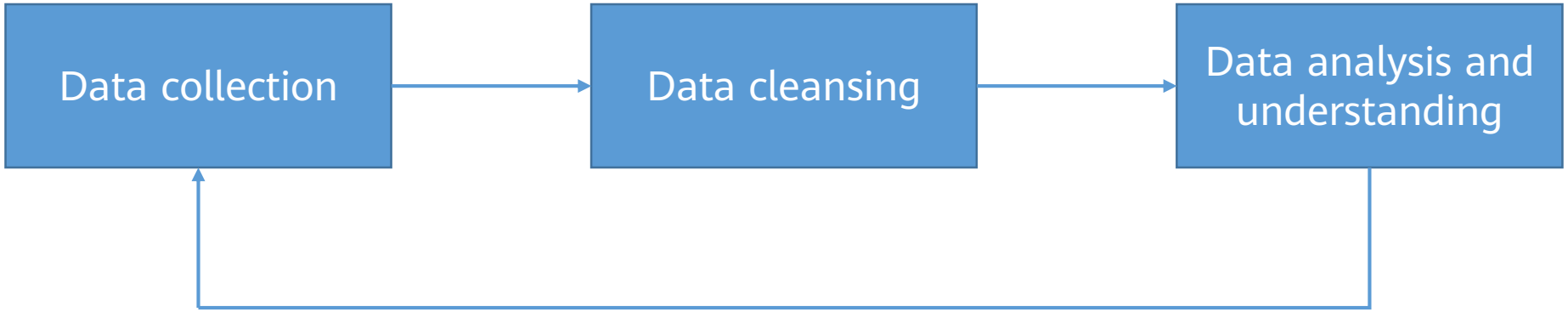
Data Construction

- Build datasets based on application scenarios and feature factors, including but not limited to data types, factor proportion, environment factors, and specific requirements. When building a dataset in the test phase, collect as much data as possible from actual situations.
- The dataset building process includes data collection, data cleansing, and data analysis and understanding.



Data Construction – Data Collection

- 1. Cold start data collection
  - (1) Collect historical data offline.
  - (2) Obtain network data using crawlers in compliance with laws and regulations.
  - (3) Purchase data from third-party data companies.
- 2. Online data backflow
  - After AI applications are published, collect data from real application scenarios in the service backend in compliance with laws and regulations.

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Data Construction – Data Cleansing

Original text:	[Event – Prize Announcement] [Huawei Cloud BBS] Are you ready for the gifts in October, Huawei Sound speakers? \(≧▽≦)/\r\nSame activities, different prizes\r\n
Clean text:	Event | Prize Announcement | Huawei Cloud BBS | Are you ready for the gifts in October, Huawei Sound speakers? Same activities, different prizes

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Data Construction – Data Analysis and Understanding

- **Analyze, understand, and select proper data for model training and iteration is important for AI application development. Understand data by performing the following steps:**
  - Sample distribution analysis: analyzes sample distribution and determines whether the constructed dataset covers comprehensive data.
  - Negative sample analysis: analyzes samples with poor test performance and finds feature factors of AI models to optimize the models.

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Data Construction – Negative Sample Analysis

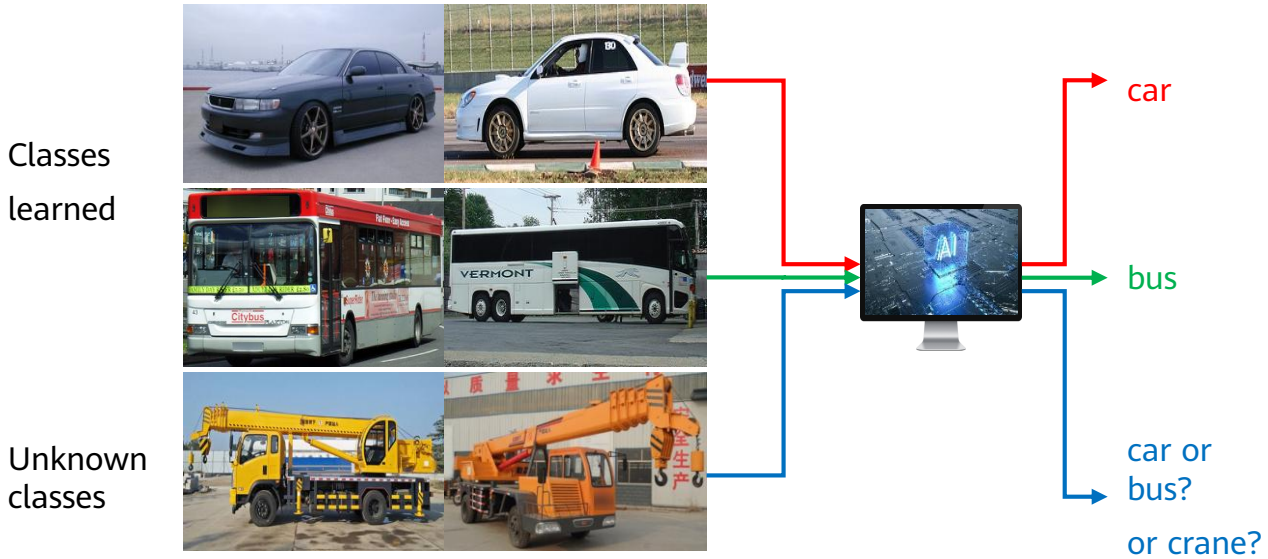
- Samples with poor test performance are classified into **hard samples** and **samples of unknown classes**.

Two hard sample types

Samples that can be identified by people but are difficult to be identified by AI, for example, vehicles and people in street views

Samples that are difficult to be identified by both people and AI, for example, license plate numbers in blurred images

Samples of unknown classes refer to samples that have not been learned by AI models.



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Evaluation Metrics

- Evaluation metrics are classified into precision metrics and performance metrics. Precision metrics vary in different AI technical fields, as shown in the following table.

Technical Field	Recognized Metrics
Image classification	Accuracy, recall, precision, F1 score, ROC curve, PR curve, and confusion matrix
Object detection	mAP(mean average precision)
Image segmentation	MIoU(Mean Intersection over Union)
OCR	Character precision rate (CPR), character recall rate (CRR)
Machine translation	Bilingual Evaluation Understudy (BLEU) ROUGE(Recall-Oriented Understudy for Gisting Evaluation)
Speech recognition	Word error rate (WER), sentence error rate (SER)

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Evaluation Metrics – Performance Metrics

- Regular performance metrics are as follows:
  - Frames per second (FPS): frequency (rate) at which consecutive images (frames) are captured or displayed
  - Floating point operations per second (FLOPs): number of floating point operations that can be performed by a computing entity in one second
  - GPU memory usage
  - Queries per second (QPS): number of requests an AI application API responds to per second

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Section Summary

- This course describes the main process of AI application testing, including scenario analysis, feature factor analysis, data construction, evaluation metrics, and automation solutions. Dataset construction is a key step in the entire process and must consider samples under different feature factors and scenarios a well as hard samples and unknown class samples after AI applications are brought online.

Q&A

1. Which of the following are performance evaluation metrics of AI applications? ()

A. FPS(Frames Per Second)

B. mAP(mean average precision)

C. GPU memory usage

D. QPS(Query Per Second)

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